

TB XYZ PANNER

User Manual - English Edition - Rev. 2026-06-30

Overview

TB XYZ Panner is a spatial panner that keeps conventional stereo left/right localization as the core, then adds procedural cues for height, distance, and front/back impression. The current design separates the stable PanCore from the auxiliary Procedural Cue layer.

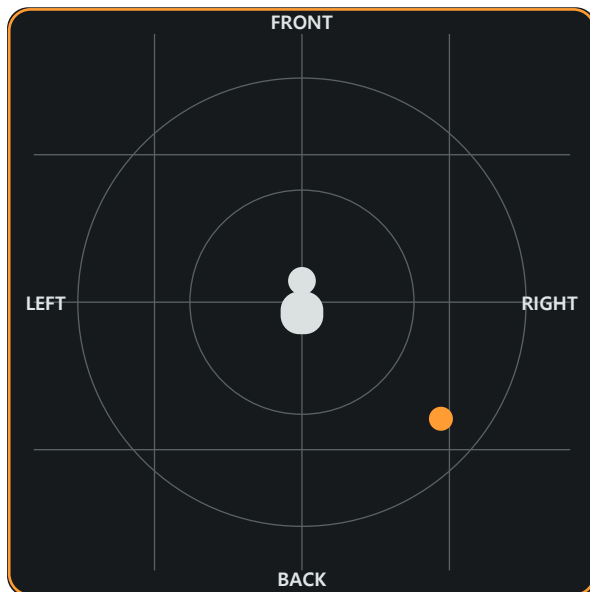
- Stereo mode prioritizes reliable left/right panning. Hard left/right is designed to close the opposite channel.
- The Top View analyzer controls Pan and Front/Back. It does not change Distance.
- The Side View analyzer controls Height and Distance.
- Headphones mode uses the included KEMAR HRIR based HRTF renderer.

Signal Structure

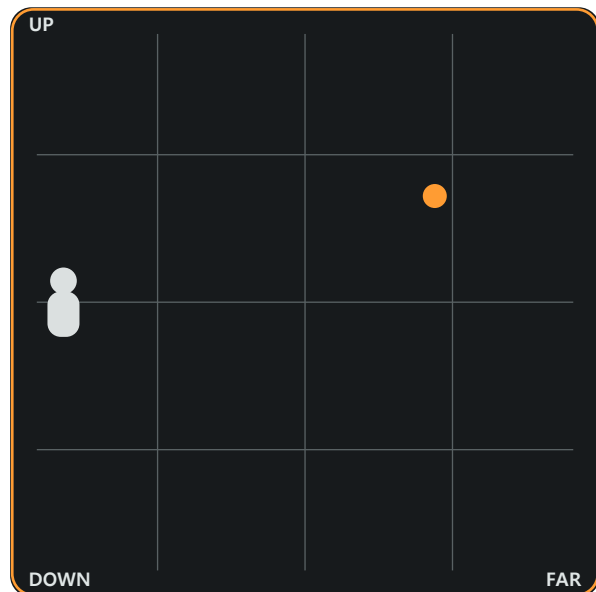
The internal model is divided into PanCore + Procedural Cue. PanCore decides the left/right position, while the cue layer adds front/back, height, distance, and room behavior.

Layer	Purpose	Important Rule
PanCore	Left/right panning and stereo input preservation	Front/back does not directly change pan gain.
Height Cue	Up/down impression	Uses high-shelf tone and ceiling/floor reflection balance.
Distance Cue	Far/near impression	Uses LPF, attenuation, and early reflection instead of moving direct delay.
Front/Back Cue	Front/rear impression	Only affects tone/reflection cues lightly; it does not rewrite the direct pan.

Interface



Top View
X axis is PAN. Y axis is FRONT/BACK. Clicking here does not alter Distance.



Side View
X axis is DISTANCE. Y axis is HEIGHT. Use it to shape depth and elevation.

Controls

Control	Range / Default	Description
PAN	-1.0 to +1.0 / 0.0	-1 is Left, 0 is Center, +1 is Right.
Front Back	-1.0 to +1.0 / +1.0	Top View Y axis. +1 is Front, -1 is Back. No dedicated knob.
HEIGHT	-1.0 to +1.0 / 0.0	Up/down cue, also controlled by the Side View Y axis.
DISTANCE	0.0 to 1.0 / 0.0	Depth cue. At 1.0, direct gain follows an approximately -12 dB curve.
ROOM	0.0 to 1.0 / 0.35	Controls early reflection amount.
OUTPUT	-24 dB to 0 dB / 0 dB	Final output trim.
STEREO	button	Stereo-compatible rendering mode. Default.
HEADPHONES	button	KEMAR HRTF based headphone rendering mode.

Basic Operation

Accurate left/right panning

- Set Render Mode to STEREO.
- Move the PAN knob or click horizontally in the Top View.
- First test hard left/right to confirm that the opposite channel closes.

Creating height and distance

- Move vertically in the Side View to change HEIGHT.
- Move right in the Side View to increase DISTANCE.
- DISTANCE is expressed through direct attenuation, low-pass damping, and early reflections. The direct path is not moved with modulated pre-delay.

Creating front/back impression

- In the Top View, up is FRONT and down is BACK.
- Front/Back does not alter PAN. Rear values are applied lightly through tone and early reflection behavior.
- On stereo speakers, rear placement should be understood as a psychoacoustic cue rather than a physically exact rear image.

Headphones mode

- Press HEADPHONES to use the KEMAR HRIR based HRTF renderer.
- The included HRIR data is a compact subset. It is not personalized and not a full 360-degree dataset, so front/back confusion may occur.

Surround

On 5.1 and 7.1 outputs, non-LFE channels are remapped using speaker azimuths. Pan rotates the bed, and rear front/back values contribute to rear azimuth calculation.

Notes and limitations

- Exact rear localization on two stereo speakers is physically limited.
- High DISTANCE and ROOM settings can make reflections more audible and soften localization.
- If the DAW keeps loading an older build, rescan the plugin or clear the host cache.
- Depending on build target settings, the Debug VST3 artifact may still appear as NewProject.vst3.

Quick Test Checklist

Test	Expected behavior
PAN = -1	Right output closes and the signal sits left.
PAN = 0	The signal sits center.
PAN = +1	Left output closes and the signal sits right.
Top View up/down	Only Front/Back changes. Distance is preserved.
Side View left/right	Distance changes.
Side View up/down	Height changes.